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FEDERAL SPECIFICATION

SCRUBBING MACHINE, FLOOR, ELECTRIC (BATTERY POWERED)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cabinet type pedestrian electric battery powered floor scrubbing machines.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and sizes. Scrubbing machines covered by this specification shall be of the disc types and of the following sizes, as specified (see 6.2):

Size 1 - 16-inch scrubbing width.
Size 2 - 21-inch scrubbing width.
Size 3 - 26-inch scrubbing width.
Size 4 - 30-inch scrubbing width.
Size 5 - 36-inch scrubbing width.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

W-B-00133 - Battery, Storage (Lead-Acid, Industrial Portable Service).
CC-M-645 - Motor, Direct Current (Fractional Horsepower).
CC-M-646 - Motor, Direct Current (Integral Horsepower, 200 HP and Smaller).
OO-P-40 - Pads, Floor (Floor Polishing, Buffing, and Scrubbing) (and Pads, Driving and Pad Holders).
OO-P-570 - Polisher, Floor, Electric (Rotary Disc).
UU-T-81 - Tags, Shipping and Stock.
PPP-B-140 - Batteries, Storage, Industrial: Preparation for Shipment and Storage of.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple Wall.
PPP-C-650 - Crates, Wood, Open and Covered.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-116 - Preservation, Methods of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications: The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Underwriters' Laboratories (UL), Inc., Standards:

UL 516 - Standard for Rectifiers.
UL 583 - Standard for Electric-Battery-Powered Industrial Trucks.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 1285 Walt Whitman Road, Melville, Long Island, NY 11749; 207 East Ohio Street, Chicago, IL 60611; or 1655 Scott Boulevard, Santa Clara, CA 95050.)

Floor and Vacuum Machinery Manufacturers Association (FVMMA), Inc., Standard:

Industry Standard No. 1 - Method of Rating Commercial and Industrial Type Vacuum Cleaners (Portable and Mobile) Suction Only.

(Application for copies should be addressed to the Floor and Vacuum Machinery Manufacturers Association, Inc., 604 Davis Street, Evanston, IL 60204.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Fire and casualty hazards.

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3.1.1 Each contractor shall submit to the contracting agency proof that the equipment he proposes to supply under this specification conforms to the requirements of UL 516 and UL 583. The UL label may be accepted as evidence that the equipment conforms to this requirement. Testing to UL 583 will be for the type E industrial truck.

3.1.2 In lieu of the label, the contractor may submit independent proof, satisfactory to the contracting agency, that the equipment conforms to the applicable requirements of the published standards including methods of tests of UL 516 and UL 583, type E.

3.1.3 Compliance with UL requirements regarding fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his equipment.

3.2 Materials. The materials used shall be suitable for the purpose intended and shall be free from defects or imperfections which may affect the appearance and serviceability of the machine. Components of the machine that are subject to wetting shall be made of corrosion resisting material or material suitably plated or coated so as to prevent rust or corrosion.

3.3 Design. The machine shall be a fully enclosed, cabinet style, key operated, battery powered, self-propelled pedestrian type. Each machine shall be new and of the manufacturer's latest design. The cabinet shall be provided with resilient, non-marking replaceable bumpers to prevent the cabinet from marring walls and baseboards.

3.4 Construction. The machine shall be of rugged construction and capable of withstanding heavy duty commercial use. The machine shall be constructed so that it can be readily disassembled for inspection, adjustment, and replacement of parts. Each part of the machine shall be interchangeable with the similar part of any other machine of the same model made by the same manufacturer.

3.5 Performance. The machine shall exhibit good performance and simplicity of operation. The brush(es) shall scrub evenly over the required scrubbing width. The squeegee shall collect all solution deposited by the brush(es) in the squeegee swath whether in a straight line or going through 90° turns. The "vacuum" shall recover all the solution collected by the squeegee. The machine shall operate smoothly and without objectionable noise. Objectionable noise is defined as sound levels above 60 dbA for institutional/hospital machines and 70 dbA for standard machines (see 4.3.2.2.3). Endurance and scrubbing performance will be in accordance with table I on smooth concrete floors.

TABLE I. Scrubbing machine sizes and performance

Size	Scrubbing width range	Machine dimensions			Tank capacities		Battery capacity (Min.)	Weight	Brush load (Min.)	Operation time continuous (Min.)	Scrubbing performance (Min.) (Ref.)
		Length (Max.)	Width (Max.)	Height (Max.)	Solution (Min.)	Recovery (Min.)		Machine less battery (Max.)			
	Ins.	Ins.	Ins.	Ins.	Gals.	Gals.	Amp-hr.	Lbs.	Lbs.	Hr.	Sq-ft/hr.
1	16-20	56	24	44	9	9	140	400	50	4	12,000
2	21-25	60	28	46	15	15	140	550	80	4	13,000
3	26-29	60	32	46	15	15	200	600	115	4	16,000
4	30-35	65	36	46	20	22	200	700	150	4	20,000
5	36-38	80	42	48	23	25	200	750	175	4	24,000

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3.6 Motors. Electric motors shall be of the heavy duty industrial type designed for operation on not less than 24 volts. Due to the lack of national standards, the motors shall comply with the intent of CC-M-645 and CC-M-646. The motors shall be continuous duty, either semi-enclosed and protected, or shall be totally enclosed. Hydraulic motors, if employed, shall be heavy duty and of good commercial quality.

3.6.1 Main drive motor. The main drive motor shall propel the machine and may, at the option of the contractor, drive the scrub brush(es). The propulsion requirements of 3.8.1 apply for the operational times specified for the size machine in table I.

3.6.2 Brush drive motor. If a separate brush drive motor is selected by the contractor, the machine shall satisfy the operational time requirements of table I for the particular size machine with both the main drive and the brush drive motor in operation.

3.6.3 Vacuum motor. The vacuum motor shall drive the vacuum producer in accordance with the requirements of 3.11 for the operational times specified for the size machine in table I.

3.7 Wheels. The machine shall be mounted on two wheels and one or two casters. The machine shall be stable. All wheels shall be bearing equipped and tired with non-marking rubber or synthetic tires. The treads of all wheels shall be grooved to eliminate tracking.

3.8 Machine propulsion. All scrubbing machines herein described shall be self-propelled. The drive mechanism shall transmit the full torque of the drive motor without slip. Speed shall be regulated by the operator through a "deadman" control located conveniently at the operator's station.

3.8.1 Propulsion speed. The self-propulsion speed of the machine shall be operator regulated and continuously variable from 0 to not less than 160 feet per minute (f.p.m.) on smooth, level, concrete floors with the brushes scrubbing at the minimum load pressure specified in table I. Maximum speed under these conditions shall be 264 f.p.m. The machine shall be capable of negotiating an 8 percent grade at no less than 88 f.p.m. while scrubbing at the specified minimum load pressures.

3.9 Controls. All controls necessary for machine operation or regulation shall be conveniently located at the operator's station.

3.9.1 Mechanical controls. All mechanical controls shall be smooth in operation and shall not require excessive force for actuation. Mechanical control positions shall be detented to prevent accidental dislodgement.

3.9.2 Electrical controls. Electrical controls and wiring shall conform to UL standards. The controls shall be electrically rated for the horsepower rating of the motor being controlled. Where thermostats or other overcurrent protective devices are employed, circuit components shall be rated no less than the trip current rating of the protective device. The scrubber shall have a key operated locking switch to prevent unauthorized use.

3.9.3 Indicators. Each machine shall be equipped with, as a minimum, a battery condition indicator. These indicators shall be located at the operator's station within the operator's normal field of vision.

3.9.4 Hydraulics. All components in the hydraulic system, if employed, shall be heavy duty and of good commercial quality. Hydraulic pistons and similar wiped components shall be resistant to the corrosive effects of the cleaning solutions and solvents employed in floor maintenance operations.

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3.10 Machine brush requirements.

3.10.1 Brushes and floor pads. Unless otherwise specified in the contract or order (see 6.2), polishing and scrubbing brushes to be used with the machine shall be in accordance with OO-P-570 and floor pads shall be in accordance with OO-P-40.

3.10.2 Brush housing. The brush housing shall afford adequate protection from the rotating brush(es). The housing (if protruding) shall be provided with resilient, non-marking replaceable bumpers. Curtains or baffles to prevent slinging the cleaning solution beyond the periphery of the housing shall be provided.

3.10.3 Brush drive. The brush(es) shall be driven by the main drive motor or by a separate motor (see 3.6.2). Brush speed shall be no less than 100 revolutions per minute (r.p.m.). The power transmission system employed shall be capable of delivering the torque required to maintain the specified r.p.m. at the maximum brush load selectable by the operator without slip.

3.10.4 Brush mounting: Machine sizes 1 and 2 shall be designed for single or twin brush operation. Twin disc machines shall have brush overlap to eliminate gaps in the cleaning swath. The brush holders shall secure the brushes but shall permit brush changing with minimum effort and without special tools. The holders shall provide for brush positioning and loading. Positioning and loading shall be controllable by the operator. When in contact with the floor, minimum brush loading shall be as specified in table I. The brush holder shall be self-aligning or articulated to permit full brush contact on uneven floors.

3.11 Vacuum producer. The vacuum producer shall be driven by a motor as specified in 3.6.3. Turbine or blade design shall be the contractor's option but the vacuum producer shall be capable of developing 6 inches water lift through a 1-1/2 inch orifice. The "vacuum" shall recover the cleaning solution collected by the squeegee and deposit it in the recovery tank. The vacuum producer exhaust shall be suitably muffled or diffused to eliminate objectionable noise. A high efficiency 0.3 micron impaction filter assembly shall be provided when specified (see 6.2). Tests shall be conducted in accordance with 4.3.2.2.6.

3.12 Squeegee. A non-marking squeegee shall be located oft of the brushes. The squeegee material shall be resilient, non-hardening, oil-resistant and impervious to common cleaning detergents and solvents. The squeegee length shall be one inch greater (minimum) than the cleaning swath. The squeegee shall be spring loaded or equipped with rollers to permit operation next to walls or protrusions. The squeegee shall be of sufficient height to prevent solution being thrown over the blade by the brush(es). The squeegee assembly shall be spring loaded to permit surface contact over uneven floors and shall be rigged to follow the brush swath. The assembly shall be capable of being raised or lowered by the operator from his station through actuation of convenient hand or foot controls.

3.13 Solution and recovery tanks. The tank capacities shall be as specified in table I. Materials, coatings, coverings, etc., shall be immune to attack by common detergents and solvents utilized in floor maintenance. Where weldments are used in tank construction, electrode or filler rod material shall approximate the base metal. The tanks shall be constructed of corrosion resistant metal when specified (see 6.2). Covers or baffles shall be provided to prevent solution from splashing out of the tanks. Synthetic coatings or finishes may be applied to the tanks and covers to provide the required corrosion protection. The tanks shall be easily accessible for cleaning and removal.

3.13.1 Clean solution tank. The clean solution tank shall be mounted in the machine so that it can be easily and rapidly filled. The flow of solution from the tank to the brush shall be controlled by means of a regulating type valve. The valve shall be controlled from the operator's position. The solution from the tank shall be fed through pipes or hoses to the center of each brush. The feed lines shall be equipped with an accessible strainer. The tank shall be provided with an auxiliary valve or similar means or method for quick emptying.

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3.13.2 Recovery tank. The recovery tank shall be provided with an automatic cutoff mechanism which will deactivate the vacuum producer system when the recovered solution reaches a predetermined level. Some means for sludge collection shall be provided in this tank; if a dump valve is employed, the valve shall permit quick emptying and shall be located no closer than 12 inches from the floor when the tank is installed in the machine. The sludge trap and/or tank shall be easily removable for cleaning. A dump valve is not required if alternate methods for tank cleaning and emptying are shown to be acceptable.

3.14 Operating handle. The machine shall be equipped with an operator's control handle. The handle shall be located at the rear of the machine and in a position convenient to the operator when walking and guiding the machine. The handle shall be provided with suitable rubber or plastic hand grips.

3.15 Battery compartment. The battery compartment shall be adequately ventilated and of ample size to house the batteries required and so constructed as to prevent movement of and damage to the batteries during scrubbing and transporting operations. The compartment cover shall be hinged or readily removable and when opened shall expose the entire battery. The inside surface of the cover if, of metal, shall be lined or coated with a suitable insulating material. In addition to the hinged or removeable cover, one or both sides of the battery compartment may be hinged or removeable at the option of the contractor. The compartment shall be constructed so that the batteries can be readily inspected, serviced, removed, and installed.

3.16 Battery. The batteries shall conform to W-B-00133 and shall be furnished by the supplier of the machine unless otherwise specified (see 6.2). The nominal potential of the batteries shall be not less than 24 volts. Capacities required will be determined by the contractor to satisfy the performance requirements of table I but in no case shall they be less than those values shown in table I.

3.17 Battery charger. A battery charger shall be furnished with each machine unless otherwise specified (see 6.2). The charger may be furnished as an accessory or may be built in. The charger shall conform to the requirements of UL 516. The charger shall have a capacity of not less than 15 amperes. The charger shall be provided with a timer that can be set to charge the batteries automatically.

3.18 Machine sizes. Machine dimensions and weight shall be in accordance with table I for the sizes indicated.

3.19 Maintenance. Periodic lubrication and maintenance requirements shall be kept to a minimum. Provisions for lubrication shall be incorporated where required. Lubricated parts shall be adequately sealed or shielded against entry of foreign matter. Lubricant retainers shall be employed where leakage may be anticipated. The machine shall be completely lubricated prior to delivery. Preparation for use will only entail reassembly of parts removed for shipment (see section 5). Maintenance shall not require special tooling. All components and assemblies subject to wear and/or replacement shall be readily accessible.

3.19.1 Hydraulic system requirements. Periodic maintenance requirements on the hydraulic system shall be kept to a minimum. A filter shall be required between the reservoir and the pump inlet. Filtration to 10 micron is required. The filter shall be equipped with a bypass indicator. Servicing shall be limited to replacement of the filter element and fluid checks.

3.20 Finish. The scrubbing machines shall have all ferrous components finished with a durable enamel, lacquer, or other suitable form of corrosion protection. Stainless steels, glass fiber, rubber, or plastic parts do not require finishing. Colors and finishes normally applied for commercial customers will be acceptable.

3.21 Marking. Each scrubbing machine shall be marked in a permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

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3.22 Instructions. The manufacturer shall furnish with each scrubbing machine a booklet or pamphlet giving complete instructions for the operation, lubrication, adjustment, maintenance, and care of the machine. An Illustrated Parts Breakdown and a repair parts list shall also be furnished with each machine.

3.23 Workmanship. Workmanship shall be first class in every respect. The scrubbing machine shall be free from defects affecting appearance and serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection for acceptance. Inspection shall be made in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter. The inspection terms and definitions shall be interpreted in accordance with MIL-STD-109.

4.2.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the specified requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2.2 Intermediate visual and dimensional examination. The intermediate product or assemblies which cannot be checked or measured after final assembly shall be examined at various stages of manufacture for compliance with this specification. Examination will be made at an appropriate intermediate station for defects in construction, workmanship, dimensions, and material. Any deviation from specified requirements shall be classified a defect.

4.2.3 End product inspection.

4.2.3.1 Visual examination. Each completely assembled end item shall be examined for defects in finish, material, construction, workmanship, and marking. Defects shall be classified as outlined in table II for recording of inspection results and corrective action. Results shall be kept available for review by the Government inspectors.

TABLE II. Classification of defects

Examine	Defect	Classification	
		Major	Minor
Finish	Type of finish not as specified or blistered, peeled, chipped, or area of no film or thin film.		X
	Not finished where required.	X	
	Component or assembly that is painted or otherwise coated where serviceability may be affected, e.g., paint on gasket.	X	
Material	Component not fabricated of the specified material.	X	
Design	Any characteristic not in accordance with the specified requirements.	X	

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TABLE II. Classification of defects (con.)

Examine	Defect	Classification	
		Major	Minor
Construction and workmanship, general (applicable to all components and assemblies)	Part missing or not specified type or size.	X	
	Fractured, split, punctured, dented, deteriorated, bowed, malformed, or otherwise impaired affecting serviceability.	X	
	Defective or otherwise damaged but not affecting serviceability.		X
	Misplaced, loose or not in proper alignment affecting serviceability.	X	
	Sharp burr, sliver or splinter that may be injurious to personnel or may affect serviceability.	X	
	Operation omitted or not properly performed, affecting serviceability.	X	
	Operation omitted or not properly performed, but not affecting serviceability.		X
	Functioning component that requires abnormal force to operate.		X
	Not connected or joined as specified or poorly accomplished, affecting serviceability.	X	
	Any adjustable assembly that cannot be adjusted or is not properly adjusted to perform the function intended.	X	
	Any component not readily accessible for servicing where required.		X
Welding and brazing (where required)	Not type specified, missing, incomplete, burn holes, cracked, fractured, or otherwise not fused.	X	
	Slag inclusion, slight undercut, not smooth and uniform, scale or flux deposits not removed.		X
Soldering (when applicable)	Missing, not adherent or incomplete.	X	
	Not clean (flux or flux residue not removed), not smooth (surface not finished neatly), or pinholes in solder.		X
Bolts, nuts, screws, studs, and other types of threaded fasteners	Missing, broken, stripped, fractured, or loose.	X	
Electrical assembly	Wiring cut, abraded, not properly joined, loose at terminal or not inclosed in specified conduit where required.	X	
	Adequate slack not provided for wiring to relieve strain or excessive insulation stripped from wiring.		X

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TABLE II. Classification of defects (con.)

Examine	Defect	Classification	
		Major	Minor
Lubrication (where applicable)	Not lubricated where required.	X	
	Insufficiently lubricated.		X
	Not specified type lubrication fitting, affecting serviceability.	X	
	Lubrication fitting not readily accessible for servicing.		X
Assembly	Component not properly assembled or secured but not affecting serviceability e.g., lockwasher missing from metal fastener.		X
	Unit perceptibly out of square or alignment.	X	
Marking identification instructions, manual	Missing, incomplete, not legible.		X

4.2.3.2 Dimensional examination. Examination shall be made of the end item to determine compliance with dimensional requirements. Any dimension that exceeds the specified tolerance shall be classified a defect. The inspection level shall be S-2 with an Acceptable Quality Level (AQL) of 6.5 defects, expressed in defects per hundred units.

4.2.3.3 Inspection of preparation for delivery requirements. An inspection shall be made to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with table III. For examination of contents, the sample unit shall be one shipping container fully prepared for delivery and selected just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

TABLE III. Classification of preparation for delivery defects

Examine	Defect
Markings (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of container.

4.2.3.4 Sampling for operational and functional tests. A random sample of scrubbing machines shall be taken from each inspection lot in accordance with the appendix to MIL-STD-105 at inspection level S-3 for operational and functional tests of 4.3. If any machine fails any of the tests, the lot shall be rejected.

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4.3 Tests.

4.3.1 Operational tests. Each electrical and mechanical control shall be actuated to determine proper operation. The drive motor control shall be actuated to determine that the propulsion system accelerates and decelerates the machine smoothly and regularly through the entire speed range. While in motion the drive control shall be released. The machine shall decelerate smoothly to a stop and the machine shall not creep until the control is again actuated.

4.3.2 Functional tests.

4.3.2.1 Test conditions. Functional tests shall be conducted on a swept floor surface consisting of resilient tile, terrazzo, or smooth concrete. Scrubbing effectiveness shall be relative to type of flooring, type of soilage, and the chemical compounds utilized. The machine shall be equipped with suitable brushes and the solution tank filled with a proper cleaning solution.

4.3.2.2 Scrubbing performance. Scrubbing performance shall be on the basis of a one-hour test run to determine conformance to the scrubbing area requirements specified in table I. In lieu of running the test for one hour, the test may be run for not less than 20 minutes and the results extrapolated to the one hour basis. Observations to be made during the scrubbing performance test shall be as specified in 4.3.2.2.1, 4.3.2.2.2, 4.3.2.2.4, and 4.3.2.2.5.

4.3.2.2.1 Motor temperature rise. The temperature rise of the motors shall be observed for conformance to the motor nameplate requirements or the motor specifications.

4.3.2.2.2 Machine speed and maneuverability. Timed tests shall be conducted to determine that the machine satisfies the speed requirements of 3.8.1. Observation shall be made to determine that machine speeds are uniform throughout the range and accelerations and decalerations are smooth. The machine shall be negotiated through random turns at various speeds and accelerations to determine that it is easily maneuverable.

4.3.2.2.3 Noise tests. The scrubber shall have the propulsion mechanism disabled, but the main drive motor, brushes, and the vacuum producer shall be in operation. Sound levels taken on portable equipment shall be noted from different points at a 6-foot radius from the vacuum producer exhaust port. The A weighted, slow response readings are the only to be taken. Background noise readings shall be noted prior to the test. Background levels greater than 10db below the machine levels shall be discounted.

4.3.2.2.4 Solution flow and leakage. Observation shall be made to determine that the flow of cleaning solution from the clean solution tank to the brush is uniform and that the flow can be properly regulated by the operator. The clean solution tank, the recovery tank, feed lines, valves, and connections shall be examined for leakage.

4.3.2.2.5 Water pickup. The floor passed over and immediately to the rear of the machine shall be observed to determine that it is only damp and that no water or puddles remain on the floor. Determination shall be made that when solution flowing into the recovery tank reaches a predetermined level, the circuit to the vacuum producer motor is de-energized.

4.3.2.2.6 Vacuum producer. The vacuum system shall be tested for compliance with the requirements of 3.11. Method of test shall be as defined by FVMMA Industry Standard No. 1 except that the intake to the vacuum shall be coupled to the orifice box through the hose furnished with the machine. Test voltage shall correspond with the rated voltage on the motor data plate. Shortening or elimination of this ducting for test purposes is not permitted. The contractor may submit a certificate of conformance if laboratory facilities are not available. Independent laboratory testing of the vacuum system for filtration and retention of 0.3 micron and larger micro-organisms shall be performed for certification purposes for institutional or hospital use. The certificate of conformance shall synopsize the test methods and filter effectiveness.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or C, as specified (see 6.2)

5.1.1 Level A.

5.1.1.1 Disassembly. Brushes shall be removed. Any other component part vulnerable to damage during shipment or storage shall be removed. All bolts, nuts, washers, and pins removed shall be placed in a bag or envelope and secured to prevent their loss.

5.1.1.2 Matchmarking. Component parts removed and the mating parts on the scrubbing machine shall be matchmarked to facilitate reassembly. Matchmarking identification shall be on shipping tags conforming to UU-T-81.

5.1.1.3 Technical publications. Technical publications for each scrubbing machine shall be preserved and packaged in accordance with MIL-P-116, method IC-1.

5.1.1.4 Spare parts. Spare parts shall be preserved and packaged in accordance with MIL-P-116. The application of the method of preservation selected for each spare part shall not damage the mechanism, structure, or function of the part and shall provide maximum protection of the part during shipment and storage.

5.1.1.5 Brushes. Brushes shall be individually packaged in boxes conforming to PPP-B-636, class 2, W5C, style RSC.

5.1.1.6 Battery charger. Each battery charger, if not built in, shall be preserved and packaged in accordance with MIL-P-116, method IIB.

5.1.1.7 Batteries. Batteries shall be prepared for shipment and storage in accordance with PPP-B-140 as specified for level A.

5.1.2 Level B. Each scrubbing machine shall be preserved and packaged as specified in 5.1.1, except that the brushes shall be packaged in boxes conforming to PPP-B-636, type I, class 1, style RSC, the battery charger shall be packaged in accordance with MIL-P-116, method IA-14, and the batteries shall be packaged in accordance with PPP-B-140 as specified for level B.

5.1.3 Level C. Scrubbing machines, battery chargers, and batteries shall be preserved and packaged in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Each complete scrubbing machine with technical publications, spare parts, and brushes shall be packed together in a close fitting box conforming to PPP-B-621, class 2, style 2, 4, or 7 as applicable, PPP-B-640, class 2, grade A, or in a covered nailed wood crate conforming to PPP-C-650. The contents in the boxes or crate shall be adequately cushioned, blocked, braced, bolted, or strapped within the container to prevent movement and subsequent damage to the contents. Each box or crate shall be closed, sealed, or strapped in accordance with the appendix to the applicable specification.

5.2.1.1 Battery charger. The battery chargers, if not built in, shall be packed in close fitting boxes conforming to PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class 2, V3C. Each box shall be closed, sealed, and strapped in accordance with the appendix to the applicable box specification.

5.2.1.2 Batteries. Batteries shall be packed in accordance with PPP-B-140 as specified for level A.

5.2.2 Level B. Each complete scrubbing machine with technical publications, spare parts, and brushes shall be packed as specified in 5.2.1, except that the boxes shall be class 1.

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5.2.2.1 Battery charger. The battery chargers shall be packed as specified in 5.2.1.1, except that the boxes shall be class 1.

5.2.2.2 Batteries. Batteries shall be packed in accordance with PPP-B-140 as specified for level B.

5.2.3 Level C. Scrubbing machines, battery chargers, and batteries shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to any markings required by the contract or order (see 6.2), interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to any marking required by the contract or order (see 6.2), interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The floor scrubbing machines covered by this specification are intended for heavy duty floor scrubbing and cleaning.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size of scrubbing machine required (see 1.2.1).
(c) Brushes and floor pads if different (see 3.10.1).
(d) When staphicidal filter assembly is required (see 3.11).
(e) Corrosion resistant steel construction when required (see 3.13).
(f) When other than the scrubbing machine supplier is to furnish the battery (see 3.16).
(g) When a battery charger is not to be furnished (see 3.17).
(h) Level of preservation and packaging required (see 5.1).
(i) Level of packing required (see 5.2).
(j) Additional marking when required (see 5.3.1 and 5.3.2).

MILITARY INTERESTS:

DOD has waived coordination on revisions and amendments to this Federal Specification until further notice.

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-DMS
NIH

Preparing activity:

GSA-FSS

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 30 cents each.

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